Tutorial 7: Bayesian Classifiers

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Task 1: Bayes decisions

For one observation $\boldsymbol{x}$, suppose:

$$\pi_1 = 0.25, \qquad \pi_0 = 0.75,$$

and the class-conditional densities at $\boldsymbol{x}$ are:

$$p(\boldsymbol{x} \mid y = 1) = 0.18, \qquad p(\boldsymbol{x} \mid y = 0) = 0.10.$$

False-positive and false-negative costs are:

$$C_{\mathrm{FP}} = 1, \qquad C_{\mathrm{FN}} = 3.$$

1. Compute $p(y = 1 \mid \boldsymbol{x})$ and $p(y = 0 \mid \boldsymbol{x})$.
2. What is the decision under zero–one loss?
3. Compute the conditional risk of deciding class 0 and class 1. What is the minimum-risk decision?
4. Derive the posterior threshold for predicting class 1 and evaluate it numerically.

Solution 1: Posterior probabilities

1. Compute the posterior probabilities.
The two unnormalized Bayes scores are:

$$s_1 = p(\boldsymbol{x} \mid y = 1)\pi_1 = 0.18(0.25) = 0.045,$$

and:

$$s_0 = p(\boldsymbol{x} \mid y = 0)\pi_0 = 0.10(0.75) = 0.075.$$

The normalization constant is:

$$s_1 + s_0 = 0.045 + 0.075 = 0.120.$$

Therefore:

$$\boxed{p(y = 1 \mid \boldsymbol{x}) = \frac{0.045}{0.120} = 0.375}$$

and:

$$\boxed{p(y = 0 \mid \boldsymbol{x}) = 0.625}.$$

Solution 1: Zero–one decision

2. What happens under zero–one loss?
With equal error costs, choose the class with the larger posterior probability:

$$a^*(\boldsymbol{x}) = \arg\max_y p(y \mid \boldsymbol{x}).$$

Since:

$$p(y = 0 \mid \boldsymbol{x}) = 0.625 > p(y = 1 \mid \boldsymbol{x}) = 0.375,$$

the decision is:

$$\boxed{\text{class } 0}.$$

Under zero–one loss, the 0.5 posterior threshold is not crossed.

Solution 1: Conditional risk

3. Compute the two conditional risks.
If we decide class 0, the only error is a false negative:

$$\mathcal{R}(0 \mid \boldsymbol{x}) = C_{\mathrm{FN}} p(y = 1 \mid \boldsymbol{x}).$$

Thus:

$$\mathcal{R}(0 \mid \boldsymbol{x}) = 3(0.375) = \boxed{1.125}.$$

If we decide class 1, the only error is a false positive:

$$\mathcal{R}(1 \mid \boldsymbol{x}) = C_{\mathrm{FP}} p(y = 0 \mid \boldsymbol{x}).$$

Thus:

$$\mathcal{R}(1 \mid \boldsymbol{x}) = 1(0.625) = \boxed{0.625}.$$

Since $0.625 < 1.125$:

$$\boxed{\text{minimum-risk decision: class } 1}.$$

Solution 1: Decision threshold

4. Derive the posterior threshold.

Predict class 1 when:

$$C_{\mathrm{FP}} p(y = 0 \,|\, \boldsymbol{x}) < C_{\mathrm{FN}} p(y = 1 \,|\, \boldsymbol{x}).$$

Let:

$$p_1 = p(y = 1 \,|\, \boldsymbol{x}), \qquad p(y = 0 \,|\, \boldsymbol{x}) = 1 - p_1.$$

Then:

$$C_{\mathrm{FP}}(1 - p_1) < C_{\mathrm{FN}} p_1.$$

Rearranging:

$$p_1 > \frac{C_{\mathrm{FP}}}{C_{\mathrm{FP}} + C_{\mathrm{FN}}}.$$

Therefore:

$$\boxed{p_1 > \frac{1}{1+3} = \frac{1}{4} \approx 0.250}.$$

Since $0.375 > 0.250$, predict class 1.

Solution 1: Interpretation

The same posterior probabilities produce different decisions:

Decision rule	Prediction
zero–one loss	0
cost-sensitive loss	1

The probability model has not changed.
Only the decision costs changed:

$$\boxed{\text{posterior probability} \neq \text{final decision}}.$$

A large false-negative cost lowers the threshold for predicting the positive class.

Task 2: LDA boundary

Consider two Gaussian classes with:

$$\boldsymbol{\mu}_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \qquad \boldsymbol{\mu}_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix},$$

and shared covariance:

$$\boldsymbol{\Sigma} = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}.$$

Initially assume equal priors $\pi_0 = \pi_1 = 0.5$.

1. Compute $\boldsymbol{\Sigma}^{-1}$ and the LDA weight vector $\boldsymbol{w}$.
2. Compute the intercept b and write the decision boundary.
3. Classify $\boldsymbol{x}_A = (2, 1)^\mathsf{T}$ and $\boldsymbol{x}_B = (1, 0.5)^\mathsf{T}$.
4. Change the priors to $\pi_0 = 0.75$ and $\pi_1 = 0.25$. Compute the new intercept and reclassify $\boldsymbol{x}_A$.

Solution 2: Weight vector

1. Compute Σ^{-1} and w.
Since Σ is diagonal:

$$\Sigma^{-1} = \begin{bmatrix} 1/4 & 0 \\ 0 & 1 \end{bmatrix}.$$

For binary LDA:

$$\boldsymbol{w} = \Sigma^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0).$$

Here:

$$\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Therefore:

$$\boxed{\boldsymbol{w} = \begin{bmatrix} 0.75 \\ 2 \end{bmatrix}}.$$

Solution 2: Intercept

2. Compute b under equal priors.
For binary LDA:

$$b = -\frac{1}{2}\boldsymbol{\mu}_1^\mathsf{T}\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_1 + \frac{1}{2}\boldsymbol{\mu}_0^\mathsf{T}\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_0 + \log\frac{\pi_1}{\pi_0}.$$

Since $\boldsymbol{\mu}_0 = \mathbf{0}$ and $\pi_1 = \pi_0$:

$$b = -\frac{1}{2}\boldsymbol{\mu}_1^\mathsf{T}\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_1.$$

Now:

$$\boldsymbol{\mu}_1^\mathsf{T}\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_1 = 3(0.75) + 2(2) = 6.25.$$

Hence:

$$\boxed{b = -3.125}.$$

Solution 2: Decision boundary

The LDA score difference is:

$$g_1(\boldsymbol{x}) - g_0(\boldsymbol{x}) = \boldsymbol{w}^\mathsf{T}\boldsymbol{x} + b.$$

Using:

$$\boldsymbol{w} = \begin{bmatrix} 0.75 \\ 2 \end{bmatrix}, \qquad b = -3.125,$$

we obtain:

$$g_1(\boldsymbol{x}) - g_0(\boldsymbol{x}) = 0.75x_1 + 2x_2 - 3.125.$$

The decision boundary is therefore:

$$\boxed{0.75x_1 + 2x_2 - 3.125 = 0}.$$

Predict class 1 when the score is positive.

Solution 2: Test points

3. Classify $\boldsymbol{x}_A$ and $\boldsymbol{x}_B$.

For:

$$\boldsymbol{x}_A = (2, 1)^\mathsf{T},$$

the score is:

$$0.75(2) + 2(1) - 3.125 = 0.375.$$

Thus:

$$\boxed{\boldsymbol{x}_A \rightarrow \text{class } 1}.$$

For:

$$\boldsymbol{x}_B = (1, 0.5)^\mathsf{T},$$

the score is:

$$0.75(1) + 2(0.5) - 3.125 = -1.375.$$

Thus:

$$\boxed{\boldsymbol{x}_B \rightarrow \text{class } 0}.$$

Solution 2: Unequal priors

4. Change to $\pi_0 = 0.75$ and $\pi_1 = 0.25$.
The weight vector does not change because the class means and covariance are unchanged.
The new intercept is:

$$b_{\text{new}} = -3.125 + \log \frac{0.25}{0.75}.$$

Since:

$$\log(1/3) \approx -1.099,$$

we obtain:

$$\boxed{b_{\text{new}} \approx -4.224}.$$

The new boundary is approximately:

$$\boxed{0.75x_1 + 2x_2 - 4.224 = 0}.$$

Solution 2: Prior effect

Reclassify:

$$\boldsymbol{x}_A = (2, 1)^{\mathsf{T}}.$$

The new score is:

$$0.75(2) + 2(1) - 4.224 = -0.724.$$

Therefore:

$$\boxed{\boldsymbol{x}_A \rightarrow \text{class } 0}.$$

Previously, with equal priors:

$$\boldsymbol{x}_A \rightarrow \text{class } 1.$$

Making class 1 rarer shifts the boundary toward class 1:

$$\boxed{\text{rarer class} \rightarrow \text{stronger feature evidence required}}.$$

Task 3: Naive Bayes

A binary training set contains 10 examples.
For class 1:

$$N_1 = 6, \qquad N_{11} = 5, \qquad N_{21} = 6.$$

For class 0:

$$N_0 = 4, \qquad N_{10} = 2, \qquad N_{20} = 1.$$

Here N_{jk} counts class-k examples with binary feature $x_j = 1$.
Use Bernoulli Naive Bayes with Laplace smoothing $\alpha = 1$.
For the test point $\boldsymbol{x} = (1, 0)$:

1. Compute the class priors and all four smoothed feature probabilities.
2. Compute the class-conditional likelihoods and unnormalized class scores.
3. Normalize the scores to obtain the posterior probabilities.
4. Give the zero–one-loss prediction.
5. With $C_{\mathrm{FN}} = 4$ and $C_{\mathrm{FP}} = 1$, give the minimum-risk prediction.
6. What would happen to the class-1 likelihood without smoothing?

Solution 3: Priors

1. Compute the class priors.
There are 10 training observations:

$$N_1 = 6, \qquad N_0 = 4.$$

Therefore:

$$\boxed{\pi_1 = \frac{6}{10} = 0.6}$$

and:

$$\boxed{\pi_0 = \frac{4}{10} = 0.4}.$$

These priors represent the observed class frequencies.

Solution 3: Smoothed probabilities

For Bernoulli features with $\alpha = 1$:

$$\hat{\theta}_{jk} = \frac{N_{jk} + 1}{N_k + 2}.$$

For class 1:

$$\hat{\theta}_{11} = \frac{5+1}{6+2} = \boxed{\frac{3}{4}},$$

and:

$$\hat{\theta}_{21} = \frac{6+1}{6+2} = \boxed{\frac{7}{8}}.$$

For class 0:

$$\hat{\theta}_{10} = \frac{2+1}{4+2} = \boxed{\frac{1}{2}},$$

and:

$$\hat{\theta}_{20} = \frac{1+1}{4+2} = \boxed{\frac{1}{3}}.$$

Solution 3: Class likelihoods

2. Compute the likelihoods for $\boldsymbol{x} = (1, 0)$.

For class 1:

$$p(\boldsymbol{x} \mid y = 1) = \hat{\theta}_{11}(1 - \hat{\theta}_{21}).$$

Thus:

$$p(\boldsymbol{x} \mid y = 1) = \frac{3}{4}\left(1 - \frac{7}{8}\right) = \frac{3}{4}\frac{1}{8} = \boxed{\frac{3}{32}}.$$

For class 0:

$$p(\boldsymbol{x} \mid y = 0) = \hat{\theta}_{10}(1 - \hat{\theta}_{20}).$$

Thus:

$$p(\boldsymbol{x} \mid y = 0) = \frac{1}{2}\left(1 - \frac{1}{3}\right) = \boxed{\frac{1}{3}}.$$

Solution 3: Class scores

Multiply each likelihood by its prior.
For class 1:

$$s_1 = \pi_1 p(\boldsymbol{x} \mid y = 1) = 0.6 \left(\frac{3}{32}\right) = \boxed{\frac{9}{160}} \approx 0.0563.$$

For class 0:

$$s_0 = \pi_0 p(\boldsymbol{x} \mid y = 0) = 0.4 \left(\frac{1}{3}\right) = \boxed{\frac{2}{15}} \approx 0.1333.$$

The class-0 score is larger before normalization.

Solution 3: Posterior probabilities

3. Normalize the two scores.

The total score is:

$$s_1 + s_0 = \frac{9}{160} + \frac{2}{15} = \frac{91}{480}.$$

Therefore:

$$p(y = 1 \mid \boldsymbol{x}) = \frac{s_1}{s_1 + s_0} = \boxed{\frac{27}{91}} \approx 0.297.$$

Similarly:

$$p(y = 0 \mid \boldsymbol{x}) = \boxed{\frac{64}{91}} \approx 0.703.$$

The posterior probabilities sum to one.

Solution 3: Zero–one decision

4. Give the zero–one-loss prediction.
Under equal misclassification costs, choose the larger posterior.
Since:

$$p(y = 0 \mid \boldsymbol{x}) \approx 0.703 > p(y = 1 \mid \boldsymbol{x}) \approx 0.297,$$

the prediction is:

$$\boxed{\text{class } 0}.$$

Equivalently, the class-1 posterior is below the 0.5 threshold.

Solution 3: Cost-sensitive decision

5. Use $C_{\mathrm{FN}} = 4$ and $C_{\mathrm{FP}} = 1$.
The class-1 threshold is:

$$\frac{C_{\mathrm{FP}}}{C_{\mathrm{FP}} + C_{\mathrm{FN}}} = \frac{1}{1+4} = \boxed{0.20}.$$

Since:

$$p(y = 1 \mid \boldsymbol{x}) \approx 0.297 > 0.20,$$

the minimum-risk prediction is:

$$\boxed{\text{class } 1}.$$

Again, unequal costs change the decision without changing the fitted Naive Bayes model.

Solution 3: Risk check

The same result can be verified directly from conditional risk.
If we decide class 0:

$$\mathcal{R}(0 \mid \boldsymbol{x}) = 4\left(\frac{27}{91}\right) = \boxed{\frac{108}{91}} \approx 1.187.$$

If we decide class 1:

$$\mathcal{R}(1 \mid \boldsymbol{x}) = 1\left(\frac{64}{91}\right) = \boxed{\frac{64}{91}} \approx 0.703.$$

Therefore:

$$\mathcal{R}(1 \mid \boldsymbol{x}) < \mathcal{R}(0 \mid \boldsymbol{x}),$$

confirming:

$$\boxed{\text{choose class } 1}.$$

Solution 3: Why smoothing matters

6. What happens without smoothing?

For class 1, feature $x_2 = 1$ appears in all six training examples.
Without smoothing:

$$\hat{\theta}_{21} = \frac{6}{6} = 1.$$

But the test observation has:

$$x_2 = 0.$$

Therefore:

$$p(x_2 = 0 \,|\, y = 1) = 1 - \hat{\theta}_{21} = 0.$$

Hence:

$$\boxed{p(\boldsymbol{x} \,|\, y = 1) = 0}.$$

One unseen feature value would eliminate class 1 completely.

Solution 3: Interpretation

Laplace smoothing changes:

$$p(x_2 = 0 \mid y = 1)$$

from:

$$0$$

to:

$$1 - \frac{7}{8} = \frac{1}{8}.$$

This prevents one small-sample count from creating absolute certainty.
The full task therefore connects:

$$\boxed{\text{smoothed feature models} \rightarrow \text{Naive Bayes posterior} \rightarrow \text{Bayes decision}}.$$

Tutorial 7 complete

Questions and discussion